CS 6515/4540: Graduate Algorithms

Fall 2026 – Exam 2 – Section A

October 27, 2026 Time: 70 minutes Total: 100 points

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Name: _____

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INSTRUCTIONS

1. Write your name, GTID, or GT username on every page.
2. Bringing two pages of notes is permitted. No books, calculators, or other electronic devices are permitted.
3. This examination contains **8 (eight) pages**, including this cover. Verify that no pages are missing.
4. Write only on the **FRONT** of each sheet. Backs will not be scanned.
5. Make answers clear and concise. For Question 1, give a short justification unless the part says otherwise.
6. You may refer to standard algorithms and theorems from class without giving their full pseudocode or proofs, unless a part asks for a proof.
7. Parts are graded separately. You may use a stated claim from an earlier part even if you did not prove it. Show your reasoning for partial credit.
8. Unless stated otherwise, arithmetic and comparisons take $\mathcal{O}(1)$ time, arrays use indices starting at 1, and log means $\log_2$.

Question	Points
1. MCQs and Short answers	40
2. Approximation Algorithm	15
3. Convex Optimization and Gradient Descent	20
4. Deterministic Rounding	25
Total	100

I am aware of and in accordance with the Academic Honor Code of Georgia Tech and the Georgia Tech Code of Conduct. I will use no person's help on this test. I have read all the instructions on this page.

Signature: _____

1 Short Answer Questions (40 points)

Point values are shown for each part. Give a short justification (1–2 lines unless a proof or calculation is requested); a counterexample suffices for a false statement.

(a) (5 points) Consider a feasible primal maximization LP and its feasible dual minimization LP. Select ALL statements that are always true.

- ◯ The primal objective value is at most the dual objective value.
- ◯ If a primal-feasible solution and a dual-feasible solution have equal objective values, both are optimal.
- ◯ If x^* is an optimal solution of the primal, every constraint (of the primal) is tight.
- ◯ If the primal has a finite optimal value, the dual has an optimal solution with the same value.

(b) (5 points) Suppose a primal constraint holds strictly at an optimal primal solution x^*. What can we conclude about the corresponding variable in any optimal dual solution?

- ◯ It must equal zero.
- ◯ It must be strictly positive.
- ◯ It can be either positive or zero.
- ◯ The primal optimum cannot be unique.

(c) (6 points) Consider the linear program

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \le b, \end{aligned}$$

where $x \in \mathbb{R}^n$ is unrestricted in sign.

Let x^* be a feasible solution, and let S denote the set of constraints tight at x^*. Write a_i^T for row i of A.

Suppose there do **not** exist coefficients $\lambda_i \ge 0$ such that

$$c = \sum_{i \in S} \lambda_i a_i.$$

Select all statements that must be true.

- ◯ There exists a direction v such that $c^T v > 0$ and $a_i^T v \le 0$ for all $i \in S$.
- ◯ There exists a direction v and $\varepsilon > 0$ such that $x^* + \varepsilon v$ is feasible and has strictly larger objective value.
- ◯ The linear program must be unbounded.
- ◯ Every constraint not in S at x^* remains satisfied under all sufficiently small perturbations of x^*.

Question 1 continued

(d) (10 points) Consider the constrained optimization problem

$$\begin{aligned} \min \quad & f(x_1, x_2) = \frac{1}{2}(x_1^2 + 4x_2^2) - 2x_1 - 4x_2 \\ & x_1 + x_2 \leq 2, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Starting at $x^{(0)} = (0, 0)$, take one gradient descent step with $\eta = 1$. If the resulting point is infeasible, find the feasible point closest to it in Euclidean distance. Show your work.

Question 1 continued

(e) (6 points) Let $f : K \to \mathbb{R}$ be a differentiable and convex function defined on a convex domain K. Let $x \in K$ such that for any $y \in K$,

$$\langle \nabla f(x), y - x \rangle \geq 0$$

Prove that x is a global minimizer of f.

(f) (8 points) Suppose a greedy 2−approximation algorithm for Vertex Cover returns a vertex cover of size 20, and you know that the graph contains a matching of size 8.

What are the strongest lower and upper bound on the size of the minimum vertex cover that follows from this information? Briefly justify your answer.

2 Approximation Algorithm - Greedy (15 points)

Let $G = (V, E)$ be an undirected graph with maximum degree $\Delta \geq 1$. An *Independent Set (IS)* is a set $S \subseteq V$ of vertices such that no edge $e \in E$ connects to vertices in S. The goal is to find an IS of maximum size.

Consider the following greedy algorithm for Maximum Independent Set:

Algorithm 1 Greedy Independent Set

```
1: S ← ∅
2: while V(G) ≠ ∅ do
3:     Choose an arbitrary vertex v ∈ V(G)
4:     S ← S ∪ {v}
5:     Remove v and all its neighbors from G
6: end while
7: return S
```

(a) (5 points) Prove that the algorithm returns an independent set.

(b) (8 points) Prove that the algorithm is a Δ-approximation for Maximum Independent Set.

Hint: Consider how many vertices are removed in each iteration and how it bounds the size of the set the algorithm returns.

(c) (2 points) Construct a family of graphs for which the algorithm can return an independent set of size exactly OPT/Δ.

Describe the graph family and explain which choices the greedy algorithm must make.

3 Convex Optimization (20 points)

Consider running gradient descent (unconstrained, over $\mathbb{R}^n$) on a convex, β-smooth function $f : \mathbb{R}^n \to \mathbb{R}$ where $\beta > 0$. We begin at $x_0 \in \mathbb{R}^n$ and we define our update to be

$$x_{k+1} = x_k - \frac{1}{\beta}\nabla f(x_k)$$

We will show a fact that was somewhat explored in the homework: that this gradient descent will make the gradient smaller as we go on.

(a) (8 points) Show

$$||\nabla f(x_{k+1})||_2 \leq ||\nabla f(x_k)||_2$$

i.e. the gradient decreases in 2-norm.

Hint: You may use the following fact without proof: For convex, β-smooth function f and any $x, y \in \mathbb{R}^n$, we have

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{\beta}||\nabla f(x) - \nabla f(y)||_2^2$$

Hint 2: In particular, try plugging in $x = x_k$ and $y = x_{k+1} = x_k - \frac{1}{\beta}\nabla f(x_k)$

(b) (6 points) Let $x^* \in \arg\min f(x)$. Prove that

$$||\nabla f(x_k)||_2 \leq \sqrt{\frac{2\beta}{k+1}\left(f(x_0) - f(x^*)\right)}$$

Hint: You may use $\sum_{i=0}^{k} ||\nabla f(x_i)||_2^2 \leq 2\beta\left(f(x_0) - f(x^*)\right)$ without proof. (This was a homework problem!)

(c) (6 points) Fix some optimum $x^* \in \arg\min f$, and let $D = ||x_0 - x^*||_2$. Show

$$||\nabla f(x_k)||_2 \leq \frac{\beta D}{\sqrt{k+1}}$$

Hint: Use part (b) + definition of β-smooth function.

4 Approximation Algorithms (30 points)

Let $U = \{u_1, \ldots, u_n\}$ be a universe of elements, and let $S_1, \ldots, S_m \subseteq U$ be subsets whose union is U. Assume that every element belongs to **at most** k **sets**, where $k \geq 1$. The goal is to select the minimum number of sets whose union is U.

For each set S_i, introduce a variable x_i indicating whether S_i is selected.

1. (6 pts) Write an integer linear program (ILP) for this problem.

2. (6 pts) Let x^* be an optimal solution to the LP relaxation. Consider the rounding rule

$$\widehat{x}_i = \begin{cases} 1 & \text{if } x_i^* \geq 1/k, \\ 0 & \text{otherwise.} \end{cases}$$

Prove that the collection $\{S_i \mid \hat{x}_i = 1\}$ covers U.

3. (8 pts) Prove that the rounding algorithm is a k-approximation.

4. (5 pts) Construct an instance with $k = 3$ and an optimal fractional LP solution x^* for which the rounding algorithm selects three sets, while an optimal integral solution selects only one set.

   Specify the universe, the sets, and the fractional solution.

END OF EXAM